

Exploring the Accuracy of NFL Betting Lines

Introduction

This paper will examine relationship between betting line accuracy and team market size in NFL games. Specifically, I will measure the effect of TV market sizes on the difference between the predicted and actual spread for a given NFL game. With this, we can potentially discern whether spread predictions are more accurate for larger market games. This paper will not focus on potential causality between market sizes and spread predictions, as there are many other factors that determine accuracy, but rather examine the relationship between them.

This question is interesting because it illuminates whether sportsbooks prioritize accuracy for games they know will receive more bets. An oddsmaker stands to benefit more from having an accurate prediction on a high-profile game because minimizing losses on a higher total volume of bets will be more profitable than doing so on a lower profile one. Most of the profit for a sportsbook comes from the commission charged for each bet, but inaccurate odds can lead to the payout from the losers of the bet being less than the cost to pay the winners of the bet, forcing the sportsbook to compensate for the difference. Thus, it is in the best interest of the oddsmaker to write odds that are most likely to minimize this compensation.

Figure 1

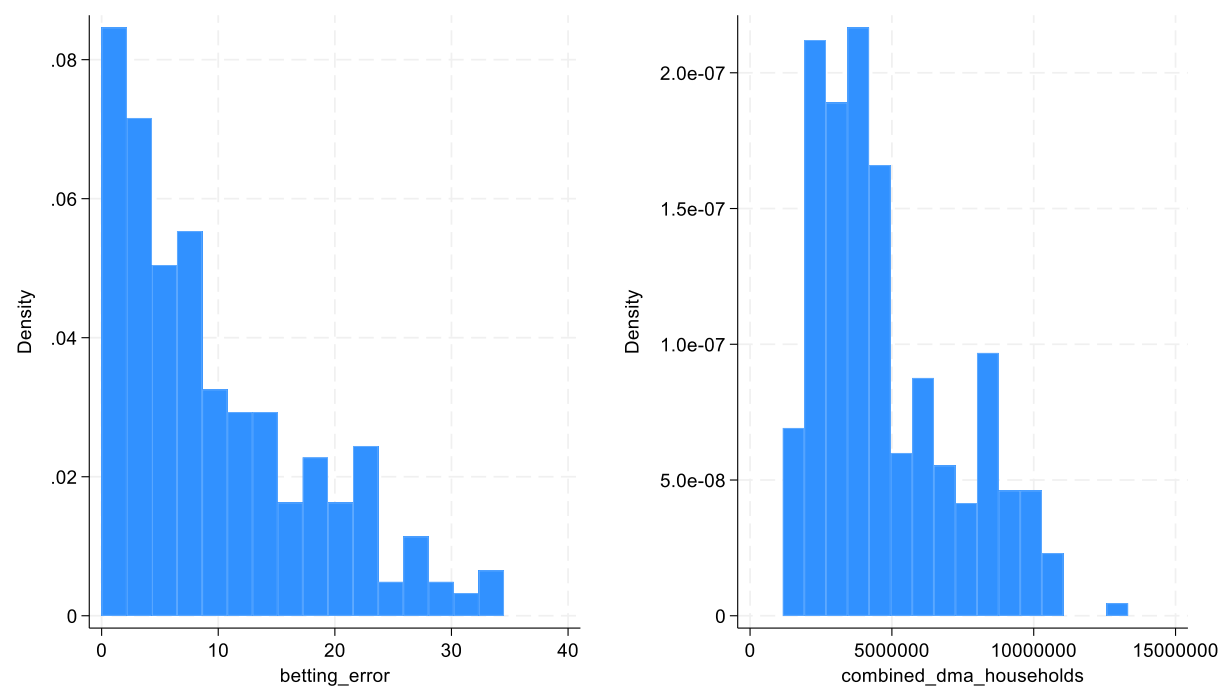
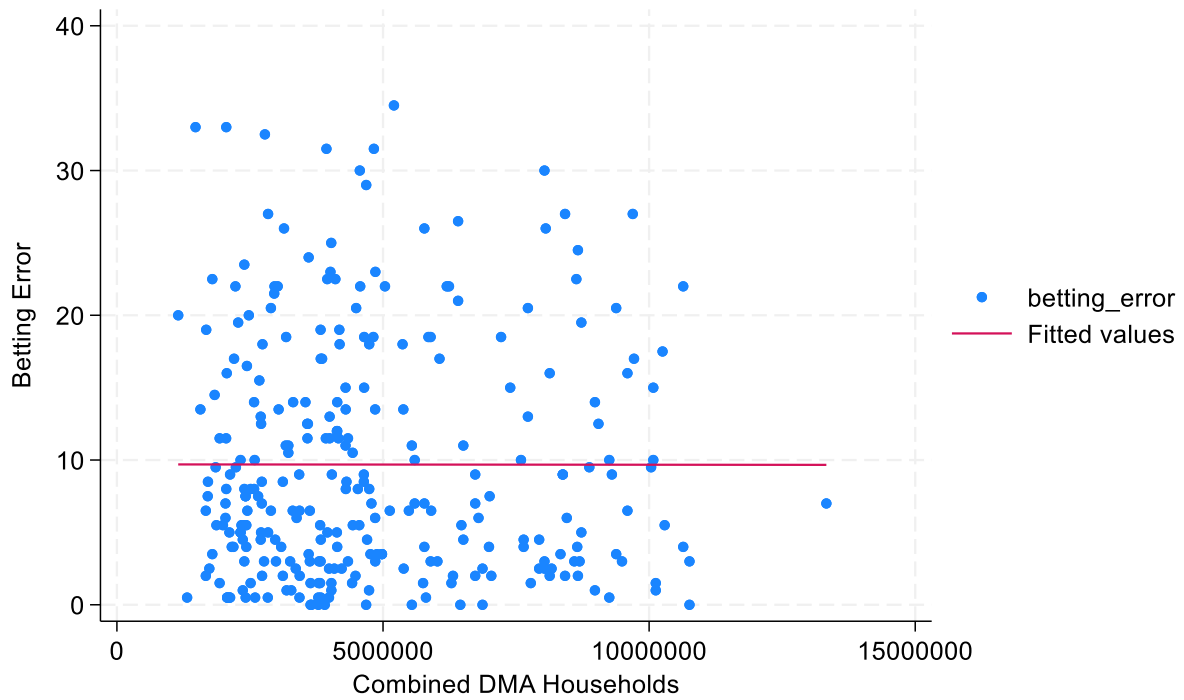


Table 1

Variable	Obs	Mean	Std. dev.	Min	Max
betting_error	285	9.689474	8.212252	0	34.5
combined_dma_households	285	4814227	2482750	1151760	1.33e+07
schedule_week	285	9.950877	5.611015	1	22
playoff_num	285	.045614	.2090136	0	1

Correlation = -0.0007

Figure 2



Regression Equation

$$\text{Betting Error} = \beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + u$$

β_1 is the coefficient of interest. This will show the average observed change in betting accuracy for an increase in one DMA household.

Table 2

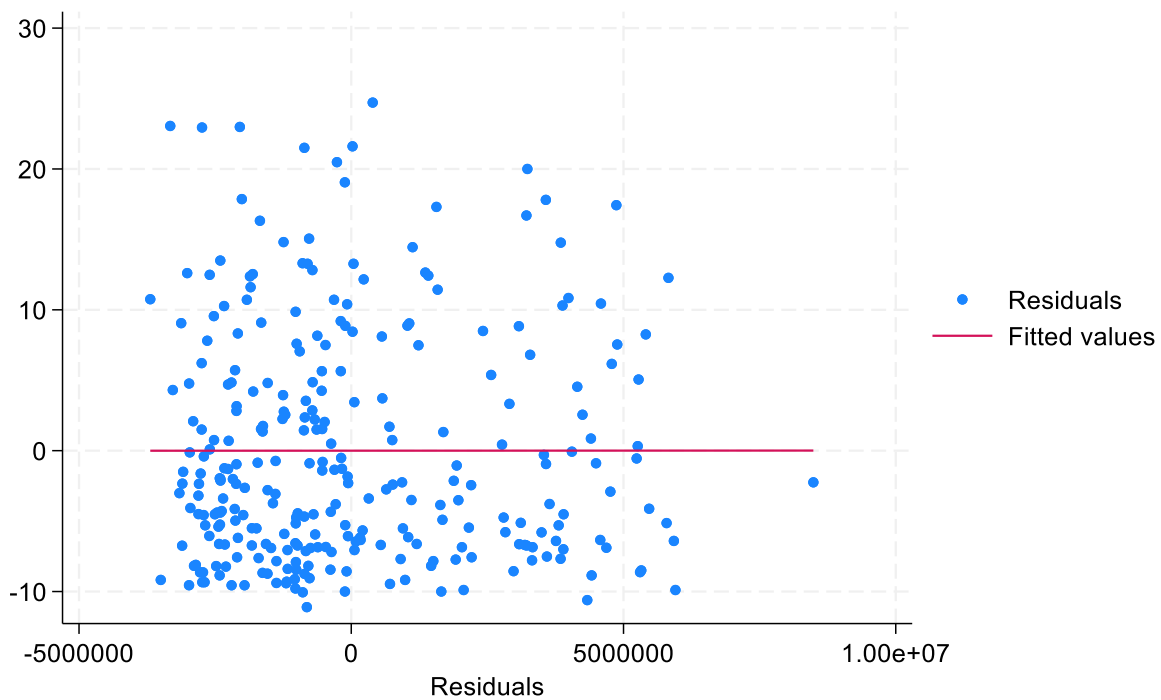
Linear regression		Number of obs	=	285		
		F(3, 281)	=	0.32		
		Prob > F	=	0.8125		
		R-squared	=	0.0038		
		Root MSE	=	8.2403		
betting_error	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
combined_dma_households	8.38e-10	1.91e-07	0.00	0.997	-3.75e-07	3.77e-07
schedule_week	-.0538878	.0969445	-0.56	0.579	-.2447176	.1369419
playoff_num	2.576826	2.746309	0.94	0.349	-2.829125	7.982777
_cons	10.10413	1.515329	6.67	0.000	7.121292	13.08697

We observe that for an increase in 1 DMA household for a given game, on average there is approximately no associated change in betting accuracy, controlling for the week in the schedule and the playoff status of a game. The coefficient is not economically meaningful as it is essentially zero. This result is statistically insignificant, with a p value of 0.997. Even when scaling the combined DMA households variable to be in 10000s instead of single households, the coefficient is still trivial. Accordingly, we see that an increase in combined DMA households of 1 standard deviation results in approximately no change in betting accuracy.

$$R^2 = 0.0038$$

We observe that 0.38% of variation in betting accuracy is accounted for by variation in number of DMA households, week number of the schedule, and playoff status.

Figure 3



Omitted Variables

For this regression, omitted variables are unlikely to occur because betting odds are specifically designed to be able to adjust if there is a way to predict when they fail. In this way, they should be equally accurate under all circumstances. However, since these odds are not perfectly accurate all the time, there can be some potential variables that are correlated with their accuracy.

Weather can potentially be one of these variables. Let us suppose that milder weather is correlated with larger market sizes as larger cities tend to have less extreme weather. For example, Buffalo and Green Bay are the two smallest markets and are both prone to extreme cold in the winter months. We can create a variable to represent extremeness of

weather (which could include heat, cold, wind, etc.), which could plausibly affect betting accuracy by introducing more chaos and unpredictability into games. In this situation, extremeness of weather would be negatively correlated with market size and positively correlated with betting error. This would create an upward bias on the coefficient for market size, as a higher market size would artificially be associated with less betting error.

Extensions

Table 3

Modification	Equation
<i>Adding a control: Over/Under</i>	Betting Error = $\beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + \beta_4 \text{OverUnder} + u$
<i>Changing a variable: measuring Betting Error as a percentage of total points</i>	Betting Error (%) = $\beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + u$
<i>Interaction: Playoff Status and Market Size</i>	Betting Error = $\beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + \beta_4 \text{PlayoffGame} \times \text{CombinedDMAHouseholds} + u$
<i>Exploring a subsample: outdoor games</i>	Betting Error = $\beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + u$

Adding a control: Over/Under

$$\text{Betting Error} = \beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + \beta_4 \text{OverUnder} + u$$

The variable OverUnder in this case represents the over/under line set by oddsmakers for a given NFL game (total points scored in the game). The idea here is that as total points scored increases the chance of making a scoring prediction error may also increase.

betting_error	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
combined_dma_households	-2.45e-08	1.93e-07	-0.13	0.899	-4.03e-07	3.55e-07
schedule_week	-.0508661	.0972576	-0.52	0.601	-.2423151	.1405828
playoff_num	2.863556	2.798543	1.02	0.307	-2.645299	8.37241
over_under_line	-.1015942	.1247975	-0.81	0.416	-.3472545	.1440662
_cons	14.70458	5.637577	2.61	0.010	3.607169	25.802

$$\beta_4 = -0.101 \quad p = 0.416$$

Given this result, we do not have enough evidence to conclude that over/under has an association with betting error. Even in games with a higher predicted level of scoring, the independent variables do not have an association with accuracy.

Changing a variable: measuring Betting Error as a percentage of total points

$$\text{Betting Error (\%)} = \beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + u$$

We will modify the betting error variable to, instead of being the difference between winning margin and predicted margin in points, be a percentage of the total points scored in that game. This will effectively weight the values such that games with higher points

scored are more forgiving to the betting error. I will use actual points scored instead of predicted points scored because actual points scored reflects the true reality of the level of uncertainty in that game.

betting_error_weighted	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
combined_dma_households	3.69e-09	4.77e-09	0.77	0.440	-5.70e-09	1.31e-08
schedule_week	-.0017174	.0022774	-0.75	0.451	-.0062003	.0027655
playoff_num	.0295025	.0561054	0.53	0.599	-.0809376	.1399427
_cons	.2212489	.0324679	6.81	0.000	.1573377	.28516

We see that all the coefficients remain insignificantly different from 0. In higher-scoring games, betting accuracy remains consistent.

Interaction: Playoff Status and Market Size

$$\text{Betting Error} = \beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + \beta_4 \text{PlayoffGame} \times \text{CombinedDMAHouseholds} + u$$

We will add an interaction between the game being a playoff game and the market size of the teams. Theoretically, market sizes should matter significantly less in the playoffs because a larger share of viewers will be outside of the markets of the given teams. This means that for playoff games the effect of an increase in market sizes on betting error should be less than that of the regular season.

betting_error	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
combined_dma_households	-6.24e-08	1.93e-07	-0.32	0.747	-4.43e-07	3.18e-07
1.playoff_num	-4.333462	4.870386	-0.89	0.374	-13.92068	5.25376
playoff_num#c.combined_dma_households 1	1.48e-06	1.14e-06	1.30	0.196	-7.69e-07	3.73e-06
schedule_week	-.0528103	.0970467	-0.54	0.587	-.2438439	.1382234
_cons	10.39893	1.524993	6.82	0.000	7.397021	13.40083

As seen in the results, β_4 is not significantly different from 0 ($p = 0.196$). Although this value is higher than β_1 , the standard errors and the values proximity to 0 make this result trivial. Regardless of playoff status, the effect of market size is most likely the same. Indeed, when running an F test on the interaction term and the MarketSize variable, we return $p = 0.43$.

```
. test combined_dma_households c.combined_dma_households#1.playoff_num

( 1) combined_dma_households = 0
( 2) 1.playoff_num#c.combined_dma_households = 0

      F( 2, 280) =    0.85
      Prob > F =    0.4304
```

Exploring a subsample: outdoor games

For this regression, we will limit the sample to only include outdoor games. Theoretically, outdoor games being exposed to the uncertainties of weather could decrease the betting accuracy of those games.

$$\text{Betting Error} = \beta_0 + \beta_1 \text{CombinedDMAHouseholds} + \beta_2 \text{ScheduleWeek} + \beta_3 \text{PlayoffGame} + u$$

First, for curiosity, when regressing betting error only on the outdoor status of a game, we return the following result.

betting_error	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
outdoor_game	-1.079777	1.098277	-0.98	0.326	-3.241606	1.082051
_cons	10.39796	.9515548	10.93	0.000	8.524936	12.27098

The game being outdoor does not have a statistically significant association with betting error. Now for the true regression using the subsample.

betting_error	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
combined_dma_households	-8.87e-08	2.02e-07	-0.44	0.662	-4.88e-07	3.11e-07
schedule_week	.0153214	.1113718	0.14	0.891	-.2044164	.2350593
playoff_num	-2.205388	2.936984	-0.75	0.454	-8.000092	3.589315
_cons	9.694482	1.662912	5.83	0.000	6.413536	12.97543

We see that all coefficients are still insignificantly different from 0. Even in outdoor weather, with higher uncertainty, betting predictions remain unaffected by the independent variables.

Conclusion

Through our regression and extensions we observe that market sizes have no clear association with betting accuracy. Regardless of market size for a game, NFL betting lines do not change in their prediction error of the margin of victory of the predicted winner. This is not a highly surprising result, as especially in the modern era of U.S. sports, betting has become widely legalized, popularized, and subsequently sophisticated in its underlying prediction algorithms. For further extension, it could be interesting to explore how this betting accuracy has changed over time, as the popularity of the sport, availability of legal

betting, and advancement of technology have all changed drastically over the decades-long lifespan of the NFL. Regardless, this result shows that oddsmakers are effective in accounting for external factors in setting their betting lines. If any factor were to have a systematic effect on the accuracy of their odds, this weakness could be exposed and reduce profit for that sportsbook. Furthermore, these algorithms are likely consistent between small and large market games, meaning that there is no decision to be made by the company regarding allocation of resources to ensure maximum prediction accuracy on certain games as opposed to others. Likewise, in our extensions, we see that even games with a higher inherent level of uncertainty, such as outdoor and high-scoring games, are not systematically different in their accuracy than regular games. In conclusion, we observe that modern sports betting algorithms are highly effective at maintaining a consistent accuracy level across varying conditions.